Mihael Cudic

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Education:

University of Oxford (UO), Balliol College, UK 2018-2023

Ph.D., Artificial Intelligence for Biomedical Sciences

- Fully funded through the **National Institutes of Health Oxford-Cambridge Scholars Program** (1 of 11 in USA)
- Recognition: **National Science Foundation Graduate Research Fellow Recipient** (1 of 2,000 in USA)

University of Florida (UF), USA 2014-2018

B.Sc., Electrical Engineering

- Graduated from the Honors Program and Summa Cum Laude (GPA: 3.97/4.0)
- Fully funded through the **Lombardi Scholars Program** (1 of 11 @ UF)
- Recognition: **Barry Goldwater Scholar** (1 of 240 in USA), UF's **Outstanding Scholar** (1 of 32 @ UF), UF's **Electric E Award** (1 of 8 @ UF's ECE Dept.), **x8** UF's **Dean's List**

Relevant Skills:

Machine Learning: Large Language Models • Generative AI • Neuroimage Analysis • Medical Image Analysis • Deep Learning • Segmentation • Auto-encoders

Genetics: Population Genetics • Genome Wide Association Studies • Polygenic Scores

Epidemiology: Causal Inference • Target Trial Emulation • Synthetic Controls • Stochastic Compartmental Models

Wetlab: Fluorescent Labelling • Transgenic Mouse Lines • Microscopy • Dissections

Research/Work Experience:

Massachusetts General Hospital (MGH), Boston, MA USA Sep.23-Present

Precision Psychiatry Postdoc & T32 Fellow in Precision and Genomic Medicine

Mentors: Dr. Jordan Smoller (Director of the Center for Precision Psych. at MGH) and Dr. Alkes Price (Harvard School of Public Health)

- Developing methods to increase the portability of polygenic scores across biobanks
- Characterized the changes in relative and absolute genetic risk across environmental contexts [3]
- Demonstrated the clinical utility of Large Language Models to infer depression severity from clinical notes [1]
- Led a sub-aim within an international consortium investigating the causal effects of COVID-19 containment policies on mental health [2]
  (*Additional Project Supervisors*: Dr. S. Bauermeister, Dr. M. Hernan, Dr. V. Patel, Dr. R. Kessler)

University of Oxford, Oxford, UK **& National Institutes of Health**, Bethesda, MD USA Aug.18-Jun.23

Computer Vision and Neuroscience Graduate Research Fellow

Mentors: Professor Alison Noble OBE (Former Director of the Inst. Of BioMed. Eng. at OU) and Dr. Jeffrey Diamond (Scientific Director at Nat. Inst. Of Neuro. Diseases and Stroke)

- Developed custom deep learning pipelines leveraging generative AI to accurately analyze noisy 3D images of neurons with minimal labeled ground truth data [4-5]

University of Florida, Gainesville, FL USA Jan.15-May.18

Undergraduate Researcher & University Scholar

Mentors: Dr. Jose Principe (Professor at UF)

Developed neuro-inspired [8] and kernel-based [6] computer vision algorithms for small-scale image classification

Carnegie Mellon University, Pittsburgh, PA USA Jun.-Aug.17

Machine Learning Intern & Summer Scholar

- Predicted plant genotypes using AI-extracted phenotypes from in situ images [7]

Appointments:

- **Adjunct Dunlevie Honors Professor** at UF (**4.8/5.0** evaluation for Signature Course) Jan.-May24
- Undergraduate **Engineering Computation Teaching Assistant** at UO Oct.-Ex.Dec.22
- Graduate **Machine Learning Teaching Assistant** at UF Aug.-Dec.17
- Undergraduate **Electronic Circuits Teaching Assistant** at UF Jan.-May.16

Extracurriculars & Other:

Study Abroad:

- **University of Pretoria**, South Africa: *South African Politics & History* (GPA: 4.0/4.0) May-Jun.17
- **Universidad Autónoma de Yucatán**, Mexico: *Tropical Ecology & Sustainability* (GPA: 4.0/4.0) Jun.-Aug.14

Other Roles: UK's **Young Entrepreneurs Society Member** (2022); Balliol Graduate **Returning Officer** (2022); Balliol **Football Captain** (2021-22); Balliol **Tennis Team Member** (2022); OU's **Scandinavian Society Member** (2022); **Balliol Choir Member** (2019-20); UF's **Culinary Arts Student Union's Logistic Director** (2015-16)

Selected Presentations:

2026 Boston; **Center for Genomic Medicine @ MGH**; "Putting Polygenic Scores in Context: How Intersectional Factors Affect Relative and Absolute Genetic Risk"; (Oral)

2025 Cancun, Mexico; **World Congress for Psychiatric Genetics**; "Putting Polygenic Scores in Context: How Intersectional Factors Affect Relative and Absolute Genetic Risk"; (**Oral Symposium**)

2025 Boston; **Primer on Medical and Population Genetics @ the Broad Institute**, "From Code to Clinic; A History of AI in Medicine" (Oral)

2024 Singapore; **World Congress for Psychiatric Genetics**, "Polygenic Scores across Environmental Contexts: How Intersectional Factors Affect Predictive Accuracy for Psychiatric Conditions" (Poster)

- **Early Career Investigator Program Awardee; Best Poster Presentation Winner**

2022 Oxford, UK; **Medical Image Analysis Seminar Series @ University of Oxford**, "Generative Adversarial Networks for Microscopy Images" (Oral)

2021 Online; **Sensory Neuroscience Seminar Series @ National Institute of Health**, "Using Machine Learning to Study Starburst Amacrine Cells" (Oral)

2019 Berlin, Germany; **Max Planck / HHMI Connectomics Meeting**, "Automatic Segmentation of Overlapping Neurons Using Convolutional Neural Networks" (Oral)

Selected Publications:

[1] **M. Cudic***, W. Meyerson*, et al "Multi-Criteria Validation of LLM-Inferred Depression Severity from Outpatient Psychiatry Notes", *preprint*.

[2] **M. Cudic***, J. Del La Hoz*, et al "The Impact of Extended COVID-19 Containment Policies on Psychological Distress in the United States: A Causal Investigation", *under Review at Nature Communications*.

[3] **M. Cudic**, J. Tubbs, T. Ge, J. Smoller "Putting Polygenic Scores in Context: How Intersectional Factors Affect Relative and Absolute Genetic Risk", *accepted to American Journal for Human Genetics*.

[4] **M. Cudic**, J. Diamond, A. Noble "Neuron Segmentation from Low Signal-to-Noise 3D Fluorescent Microscopy Images using Local and Global Scale Fusion," *Fluorescence Imaging of the Brain*, 59.

[5] **M. Cudic**, J. Diamond, A. Noble "Unpaired Mesh-to-Image Translation for 3D Fluorescent Images of Neurons," *Medical Image Analysis,* (2023), 102768.

[6] **M. Cudic**, J. Principe, "Using a Recurrent Kernel Learning Machine for Small-Sample Image Classification," *Jt. Conf. Neural. Netw. (IJCNN)*, Anchorage, AK, (2019) pp. 1-6.

[7] **M. Cudic**, H. Baweja, T. Parhar, S. Nuske, "Prediction of *Sorghum bicolor* Genotype from In-situ Images Using Autoencoder-identified SNPs", *17th IEEE Intl. Conf. Machine Learn. Appl. (ICMLA)*, Orlando, FL, (2018), pp. 23-31.

[8] R. Burt, **M. Cudic** and J. Principe, "Fusing Attention with Visual Question Answering," Jt. Conf. Neural. Netw. (IJCNN), Anchorage, AK, (2017) pp. 949-953.